

Section 0 – Title and contact Information

- Title of the proposed solution track: **Barley Enriched Balady Bread for a Climate-Resilient Approach to Nutrition and Food Sufficiency in Egypt**
- Title of the AoW2 Flagship that the solution track aligns with: **Flagship 3.1 (Catalyzing market demand, policy action, and behavior change for diversified and nutritious diets)**
- Lead Scientist(s) / Contact Person(s) / Center(s): Miguel Sanchez-Garcia, Rhiannon Crichton, ICARDA
- Emails of the Lead Scientist(s) / Contact Person(s) / Center(s): m.sanchez-garcia@cgiar.org, r.crichton@cgiar.org
- Key CGIAR Center Partners (in the case of multi-Center Eols)
- Technical link to other Science Programs: B4T and BDN

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: To establish what the innovation is, who the customers and end-users are, why it matters, and how it connects to S4I's Flagships and Grand Challenges.

***The problem to be solved:** Describe the challenge you are seeking to overcome through the proposed Scaling Solution track. What country / countries will the Solution Track operate in? Be sure to clarify who you are seeking to benefit with this work, with attention paid to gender and social inclusion. (~150 words)*

Egypt faces a persistent food security challenge due to its heavy reliance on wheat imports, which cover around 60% of national demand and make it the world's largest wheat importer. This dependency imposes a massive fiscal burden, with bread subsidies alone exceeding EGP 47 billion annually (just under USD 1 billion), while exposing the system to volatile global markets and foreign currency shortages. Rapid population growth, scarce arable land, and water stress further constrain domestic wheat self-sufficiency. The proposed SST targets both rural farmers and urban/subsidized bread consumers in Egypt—integrating barley flour into Balady bread production—addresses these challenges by reducing import dependence on wheat, lowering subsidy costs, and enhancing nutritional quality. Significantly, the inclusion of barley flour increases water absorption during dough preparation, allowing bakeries to produce more loaves from the same quantity of flour. This improves efficiency within the subsidy system and expands bread availability for low-income households, particularly benefiting women who manage household food security and smallholder farmers in marginal lands who gain from expanded barley markets.

***Your scaling ambition:** Describe the ambition of your proposed Scaling Solution track. Which of the KPIs do you believe you will contribute most towards, and at what level of quantitative and qualitative impact? (~150 words)*

Our scaling ambition is to integrate wheat–barley flour blends into Egypt's national Balady bread system, achieving measurable reductions in dependence on wheat imports and having a direct efficiency gain within Egypt's massive subsidy program, while lowering production costs and increasing availability for consumers. At scale, the intervention could save 1–2 million tons of

wheat annually, reduce import expenditures by many millions of USD, providing more bread to over 70 million Egyptians.

The track contributes directly to S4I KPIs by reaching ~5 million households (~10% of Balady bread consumers) with improved nutrition and dietary diversity, generating new agrifood sector jobs across barley production, milling, and distribution (contributing to the 0.25M+ jobs target), and expanding climate-resilient barley production on marginal and saline lands (~50,000–70,000 ha) in line with biodiversity- and climate-smart practices. Policy adoption and integration into the national subsidy system support systemic food system transformation, aligning with S4I outcomes for leveraged finance, improved diets, and resilient value chains, while ensuring gender-inclusive and equitable benefits for women and smallholder farmers.

Innovation & Value Proposition – Describe:

- *the innovation or bundle or package (maturity, novelty, suitability for scaling)*
- *and the benefits it delivers to institutional partners/customers and end-users (e.g., productivity, resilience, nutrition, risk reduction, institutional capacity).*
- *Can include linkages to other science programs that will collaborate and work on the innovation/bundle/package (~150 words)*

The innovation package scales technically mature wheat–barley flour blends for Balady bread at a national level. The approach is novel for being embedded within Egypt’s national subsidy and milling systems, enabling rapid, large-scale adoption. Institutional partners are MoSIT, ARC, and flour mills, and they will benefit from reduced exposure to wheat-import dependency, improved subsidy efficiency, and clearer QA and traceability, resulting in operational risk reduction. End-users—particularly low-income households and women who manage household food security with limited access—gain affordable, more nutritious bread. Smallholders in marginal/saline lands where barley grows well will gain a more stable and reliable market for their barley.

The package links to CGIAR’s Breeding for Tomorrow (B4T) to accelerate climate-resilient barley varieties tailored to food uses, and Better Diets & Nutrition to co-develop labeling, dietary guidance, and policy engagement that mainstreams the mixed-flour blends and nutrition-smart practices into Egypt’s nutrition and social protection agendas.

Customers & Alignment – Identify the main institutional customers (ministries, donor programs, agribusinesses, lenders) and end-users (farmers, SMEs, cooperatives). Show how the track aligns to an AoW2 Flagship and Grand Challenge. (~100 words)

Main institutional customers include the Ministry of Supply and Internal Trade (MoSIT), the Agricultural Research Center (ARC), donor-backed food security programs, flour-milling enterprises, bakery federations, and private traders linking farmers to markets. End-users comprise low-income consumers dependent on subsidized Balady bread, women managing household food security, smallholder farmers in marginal and saline lands, and cooperatives supplying resilient barley. The track aligns directly with AoW2 Flagship 3.1 (Catalyzing Market Demand, Policy Action, and Behavior Change for Diversified and Nutritious Diets) under Grand Challenge 3, while supporting Flagship 2.1 on scaling seed and soil fertility systems, advancing resilient, inclusive, and nutrition-smart food systems.

Evidence of Impact – Provide citations and evidence of both effectiveness (impact) and scalability in practice (uptake beyond pilots, adoption drivers). (~150 words)

Evidence from peer-reviewed trials on balady bread indicates that replacing 10–20% of wheat flour with whole barley flour enhances the nutritional quality (β -glucans, fiber, minerals) of the bread while maintaining an acceptable taste and texture. Farinograph studies indicate 9–13% higher dough water absorption at 20% barley, translating into more dough per unit of flour, enabling a higher output of loaves, so increasing efficiency within the national subsidy system. Sensory tests confirm acceptability up to 20% barley flour, consistent with other studies showing 15–30% substitution feasibility. The scalability of modified balady bread inputs is supported by Egypt's Wheat Flour Fortification Program, which successfully modified flour specifications across 143 mills, supplying ~6.5 million MT and reaching over 50 million consumers. Adoption drivers include strong alignment with MoSIT policies, established milling infrastructure, and farmer supply networks. The approach also expands the market for climate-resilient barley grown in marginal or saline lands, benefiting smallholders and strengthening local food systems while reducing Egypt's dependence on wheat imports.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Purpose: To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

Scaling Pathway & Sustainability – Describe how the innovation bundle and/or package will scale (public, private, PPP). Explain how the solution will remain viable at scale (revenues, public financing, PPP, IFI/donor support), with milestones or cost-recovery targets. (~250 words)

The proposed scaling pathway leverages a public-private partnership (PPP) model with MoSIT providing policy leadership and subsidy integration, flour mills implementing wheat–barley blending at scale, and farmers' cooperatives supplying resilient barley from reclaimed or marginal lands. Extension support from ARC and ICARDA ensures production quality and adoption of climate-smart practices, while private traders, bakeries, and distributors maintain the supply chain. Gender-inclusive measures target women smallholders and women-led bakery operations, ensuring equitable benefits.

Sustainability is embedded in multiple layers: the national bread subsidy program guarantees financing for blended flour procurement, while wheat import savings and improved loaf yield create clear cost-recovery incentives. Initial scale-up can be supported through IFI, donor, and government co-financing, covering barley seed systems, extension services, marketing campaigns, and subsidy adjustments.

Key milestones include:

1. Pilot blending in high-barley governorates with monitoring of dough performance and consumer acceptance.
2. Adoption of procurement policies prioritizing barley-wheat blends.
3. Expansion to national-level subsidy programs, covering >100 mills and ~50 million consumers.

4. Integration of resilient barley supply chains from marginal lands, reaching at least 50,000 smallholder farmers.
5. Full mainstreaming of blended flour in all Balady bakeries, achieving measurable wheat import reduction and nutritional impact.

By combining institutional leadership, market incentives, and climate-smart production, this scaling pathway ensures the innovation remains economically viable, socially inclusive, and resilient to climate pressures, while achieving large-scale food, nutrition, and livelihood impacts.

Partnerships & Leverage – *Identify internal CGIAR champions and external partners that have a stake in championing the innovation bundle or package, provide evidence of demand (surveys, requests for collaboration, alignment with policies or strategies) and commitment (MOUs, LOIs, budget allocations from partners, or other evidence of demand), and specify major non-CGIAR programs or investments this track can leverage (budgets, partners, policies, programs, etc). (~300 words)*

Internal champions include ICARDA's barley breeding and food systems research teams, particularly through the Breeding for Tomorrow (B4T) program, and Better Diets & Nutrition initiatives, providing technical support for resilient barley varieties, flour formulation, and nutrition integration. These teams collaborate closely with ARC and FTRI on varietal testing, pilot flour blends, and consumer acceptability studies.

External partners encompass the Egyptian Ministry of Supply and Internal Trade (MoSIT), the Agricultural Research Center (ARC), Food Technology Research Institute (FTRI), FAO, donor-backed programs for food security and subsidy reform, flour-milling enterprises, bakeries, farmers' cooperatives, and private traders. Evidence of demand includes multi-stakeholder workshops, government statements on wheat import reduction, bread subsidy reform priorities, and requests from MoSIT and ARC/FTRI to pilot barley–wheat blends. Commitments include ARC-FTRI pilot trials, MoSIT interest in subsidy integration, and preliminary allocation of technical and extension resources.

Leverage opportunities are substantial. The track aligns with Egypt Vision 2030, the National Climate Change Strategy 2050, and the 2030 Sustainable Agriculture Development Strategy, supporting food security, climate-smart production, and nutrition goals. It complements large-scale land reclamation programs (around 1.5 million feddans), ongoing IFI and donor investments in barley cultivation, irrigation, and CSA interventions, and the national wheat fortification program, which reaches over 50 million consumers. These frameworks provide both financial and institutional support for scaling, ensuring integration with existing public investments, policy priorities, and private-sector initiatives.

By connecting internal CGIAR expertise with committed external partners and leveraging national strategies, donor programs, and private-sector capacity, this track is well-positioned to scale barley–wheat flour blends sustainably, improving nutrition, enhancing climate resilience, and reducing dependency on wheat imports.

Internal champions include ICARDA's barley breeding and food systems research. External partners include the Agricultural Research Center (ARC), Food Technology Research Institute (FTRI), MoSIT, FAO, and donor programs targeting subsidy reform and food security. Agribusiness partners include flour mills and bakery federations. Farmer cooperatives and private traders will support local barley marketing. Demand evidence comes from national studies highlighting the fiscal unsustainability of wheat imports and bread subsidies, as well as government statements prioritizing food system resilience. Commitments include ARC-FTRI testing and MoSIT pilot interest. Leverage opportunities include Egypt Vision 2030, National Climate Change Strategy 2050, and donor investments in land reclamation and climate-smart crops. This track aligns with large-scale reclamation projects (1.5 million feddan program), offering synergy with ongoing public investments.

Risks & Bottlenecks – *Identify specific risks (financial, political, operational, social, or others) and bottlenecks (policy, institutions, finance, technology) in the specific context and place that could hinder scaling. Describe potential mitigation strategies to address these risks and bottlenecks. (~250 words)*

Scaling wheat–barley flour blends in Egypt faces multiple risks and bottlenecks across financial, political, operational, social, agronomic, institutional, and environmental dimensions.

Financial/market risks include temporary cost parity where barley flour is more expensive than wheat, FX volatility, and global wheat price shocks. Mitigation strategies include ring-fencing wheat-import savings to stabilize mill margins, introducing MoSIT price-band triggers, and allowing technically and regulator-approved substitution with other resilient flours (e.g., sorghum) when cost-effective.

Political/regulatory risks arise from altering a national staple and slow policy cycles. These are addressed by formalizing blending standards, QA protocols, and certification through MoSIT, ARC, and FTRI, coupled with early public communication to demonstrate benefits.

Operational/technology risks involve inconsistent milling, blending, and QA. Centralizing blending at mills, applying SOPs and audits, offering minor retrofit support, and digital traceability mitigate these bottlenecks.

Social and acceptance risks include perceptions of barley as “animal feed” and taste concerns. Maintaining substitution at 10–20%, coupled with labeling, nutrition messaging, and targeted engagement with women (primary household food managers), addresses these issues.

Agronomic/supply risks involve insufficient barley volumes or seed bottlenecks. Contract farming on marginal/saline lands, pre-harvest price signals, and linking to Breeding for Tomorrow (B4T) for higher-yielding, food-type barley ensure reliable supply.

Institutional capacity risks relate to coordination across MoSIT, mills, ARC/FTRI, and cooperatives. Mitigation includes an inter-agency steering group, SLAs, and monitoring frameworks to ensure efficiency, quality, and gender-inclusive reach.

Environmental risks involve crop displacement or water stress. Scaling focuses on reclaimed/marginal lands, salinity-tolerant varieties, and water-productivity benchmarks, ensuring no net displacement of priority crops.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

Purpose: *To define what can realistically be achieved with S4I support in the next 2–3 years — including achievable reach, milestones, resources, activities, and safeguards for responsible scaling.*

With S4I support (2026–2028), the track will run the process of integrating wheat–barley flour blends in 4–6 governorates, establish national QA and traceability standards (with ARC/FTRI, mills), and secure MoSIT endorsement for national rollout by Year 3. The target reach is approximately 5 million people, or around 10% of the households that rely on subsidized Balady bread, with a priority for targeting women-headed households and other vulnerable groups.

Milestones include:

Year 1: Establishing process in 4–6 barley-producing governorates; mill-level SOPs; MELIA baselines for nutrition access, efficiency gains (higher dough water absorption → more loaf yield), and farmer inclusion.

Year 2: Expansion to 8–10 governorates; finalize a national blended-flour standard; draft MoSIT circular enabling optional uptake.

Year 3: Formal policy adoption; institutionalized QA/certification; scale-ready financing tools for mill compliance and farmer procurement.

Roles and Resources:

- MoSIT: policy, procurement signals, subsidy interface.
- ARC/FTRI & ICARDA/CGIAR (B4T & Better Diets & Nutrition): standards, product/seed profiles (including naked barley), nutrition messaging, MELIA.
- Mills/bakeries/cooperatives: blending, QC, consumer outreach.

Seed and Grain Supply Scheme: Foundation seed sourced from ARC/CGIAR through B4T, leading to certified multipliers and contract farming in marginal/saline areas with forward-purchase agreements and price bands; grain QA (moisture, test weight, contaminants) aligned with the new SOPs.

Safeguards include gender-intentional targeting; environmental guardrails focusing on reclaimed/marginal lands with no displacement of priority crops.

Readiness & Responsible Scaling – Provide evidence of scaling readiness (IPSR or similar evidence). Describe actions for ensuring that scaling will be inclusive (extending to women, youth, marginalized groups) and environmentally sustainable. (~150 words)

The innovation is at a high readiness level, with feasibility trials and consumer acceptance tests already completed. Further evidence from IPSR scaling scans highlights policy alignment and institutional demand. Responsible scaling will ensure inclusion of women (as primary household food managers), youth farmers in marginal lands, and nutrition benefits for vulnerable groups. Environmental sustainability is ensured through the expansion of barley cultivation in marginal and saline areas where wheat is less viable, improving land and water productivity without displacing existing crops.

Achievable target, Resources & Activities – Define what can realistically be achieved in 2–3 years starting 2026 forward with S4I support and current/planned partnerships. Outline the key resources (staff, institutional, financial, technical) and activities (e.g., training, policy engagement, distribution) required. (~250 words)

With S4I support (2026–2028), the track will implement a sequenced, standards-driven national scale-up of wheat–barley flour blends, targeting around 5 million people (10% of current Balady bread consumers), prioritizing women-led households, youth farmers in marginal lands, and vulnerable groups.

Year 1: Implement governorate-level scale-out in 4–6 barley-producing regions, installing mill-level SOPs, QA/traceability systems (ARC/FTRI with mills), and establishing MELIA baselines on nutrition, loaf yield efficiency, and smallholder participation.

Year 2: Expand to 8–10 governorates, finalize a national blended-flour standard, and issue a MoSIT circular enabling nationwide adoption.

Year 3: Achieve mainstream adoption through MoSIT policy endorsement, institutionalized QA/certification, and scale-ready financing tools, including working-capital lines for mills and price-band/forward-purchase mechanisms for barley procurement.

Resources & Roles: MoSIT provides policy, procurement signals, and subsidy interface; ARC/FTRI & ICARDA/CGIAR (incl. Breeding for Tomorrow and Better Diets & Nutrition) manage standards, nutrition messaging, MELIA tracking, and resilient barley development; mills, bakeries, and cooperatives execute blending, QA, and consumer outreach. Seed and grain supply schemes include ARC/B4T foundation seed, certified multipliers, contract farming in marginal/saline areas, and opportunistic use of other resilient crops (e.g., sorghum) where cost-effective.

Activities: Technician training, mill audits, bakery coaching, farmer field days/seed distribution, consumer engagement campaigns, and continuous MELIA reporting to track adoption, smallholder inclusion, gender-responsive access, and efficiency improvements in loaf yield and subsidy effectiveness.

Timeline, Milestones & Monitoring – Provide an estimate of annual milestones for 2026–2028 (users reached, partnerships formed, policies adopted). Explain how progress will be monitored, lessons captured, and adaptations made. (~200 words)

2026: Scale-out in 4–6 barley-producing governorates; formalize partnerships through MOUs with MoSIT, ARC/FTRI, 6–10 mills, and bakery federations; implement mill SOPs, QA/traceability systems, and MELIA baselines capturing nutrition access, loaf yield efficiency, farmer inclusion, and gender-responsive reach. Users reached: ~1–1.5 million. Supply-side activities: multiplication of foundation seed lots and contract-farming schemes in marginal/saline areas, with initial QC pass rates $\geq 85\%$.

2027: Expand to 8–10 governorates; finalize national blended-flour standard; issue MoSIT circular enabling full adoption; scale farmer contracts and certified seed multiplication; launch consumer communication campaigns. Users reached: ~3–3.5 million. Partnerships expand by 5–8 mills/cooperatives; financial mechanisms (working-capital lines, forward-purchase agreements) fully operational.

2028: Nationwide adoption supported by MoSIT policy; institutionalized QA/certification and compliance audits; deployment of scale-ready financing tools; opportunistic integration of other resilient flours (e.g., sorghum) where market conditions favor substitution without quality loss. Users reached: ~5 million (~10% of Balady bread beneficiaries).

Monitoring and adaptive management: KPIs are tracked via the MELIA dashboard, including nutrition, loaf efficiency, smallholder market integration, gender inclusion, QA compliance, and policy milestones. Quarterly multi-stakeholder reviews inform adaptive actions, and annual learning briefs document lessons to guide national scale-up and refinement.